ABUBAKAR SADDIQUE

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience in Generative AI, RAG pipelines, LLM applications, and MLOps through internships and real-world projects. Experienced in building AI solutions using LangChain, LangGraph, FastAPI, and CrewAI. Passionate about developing production-ready AI systems that solve real business problems.

TECHNICAL SKILLS

Languages & Data Tools: Python, SQL, Pandas, NumPy

Generative AI & LLMs: LangChain, LangGraph, CrewAI, Agno, Ollama, Prompt Engineering, RAG Pipelines, AI Agents, Vector Databases, Fine-Tuning (LoRA, QLoRA)

Machine Learning & AI: Scikit-learn, XGBoost, TensorFlow, PyTorch, NLP, HuggingFace

MLOps & Deployment: Docker, FastAPI, AWS (S3, EC2, IAM), MLflow, DVC, Git

Visualization & BI: Power BI, Streamlit, Matplotlib, Seaborn, Plotly

PROFESSIONAL EXPERIENCE

Data Science Intern | Salaar Technologies Pvt. Ltd. *Oct 2025 – Apr 2026*
Lahore Pakistan

- Automated end-to-end data preprocessing pipelines in Python, including web scraping, data cleaning, and feature engineering workflows, reducing manual processing time by 15%.
- Built and evaluated supervised ML models on structured business datasets; surfaced actionable insights that directly supported data-driven decisions across two departments.
- Collaborated with cross-functional teams to integrate data science outputs into live business workflows, improving reporting accuracy and pipeline reproducibility.

Data Science Intern | DevelopersHub Corporation *Jun 2025 – Jul 2025*
Remote

- Designed interactive Power BI dashboards tracking 8+ KPIs, giving management real-time visibility into team performance and business metrics.
- Built ML classification models to extract business insights from large datasets, directly strengthening decision-support capabilities for stakeholders.
- Automated repetitive data preprocessing workflows, cutting manual effort and improving team throughput by ~15%.

FREELANCE & CONSULTING PROJECTS

Delivered 2–3 AI/ML projects for local clients and professional groups (Lahore-based networks), covering data analysis, automation scripts, and basic ML solutions. Gained experience translating client requirements into working solutions, managing timelines independently, and delivering under real-world constraints without a formal contract platform.

PROJECTS

Customer Support Assistant |RAG-based AI Chatbot | **GitHub** *Mar 2026 - Apr 2026*
Python · LangChain · FastAPI · Vector DB(pinecone) · Prompt Engineering

RAG-powered AI chatbot for an eCommerce platform delivering context-aware, accurate responses from unstructured knowledge bases.

- Designed and built a full end-to-end RAG pipeline covering document ingestion, chunk-level embedding generation, and semantic vector search using LangChain and pinecone, enabling accurate retrieval from 500+ product FAQ and policy documents.
- Engineered a scalable FastAPI backend handling real-time query routing; implemented context-aware prompt templates that reduced hallucination and improved factual accuracy by ~30% over a naive LLM baseline.
- Applied advanced retrieval strategies (MMR, metadata filtering) to optimise response relevance across multi-domain queries, resulting in consistent context-window utilisation under 80%.

AI Research Agent | **Autonomous Multi-Agent Research System** | **GitHub** *June 2026 - June 2026*

Python · CrewAI · Groq · Tavily Search API · Pydantic · Gradio

Production-grade agentic AI system that autonomously researches any topic and generates a structured markdown report using a 3-agent CrewAI pipeline.

- Designed a sequential multi-agent pipeline (Researcher → Analyst → Writer) where each agent handles a distinct responsibility web search, insight extraction, and report writing orchestrated end-to-end via CrewAI.
- Integrated Tavily Search API for real-time internet retrieval and Groq-hosted as the LLM backbone, enabling fast, cost-free inference with no OpenAI dependency.
- Implemented Pydantic validation for both input (topic length, depth constraints) and output (word count, status, timestamp), ensuring data integrity across the full pipeline.
- Built an interactive Gradio web UI with depth selector, example topics, and live markdown report rendering; reports auto-saved with timestamps to an outputs directory.

Influencer Insights | YouTube Comment Sentiment Analyzer | **GitHub** *Sep 2025 - Oct 2025*

Python · NLP · FastAPI · MLflow · Docker · AWS EC2 · Chrome Extension (JS)

Chrome extension for real-time sentiment analysis on YouTube comments, backed by a production-grade MLOps pipeline.

- Trained an NLP sentiment classification model on a 3,700+ labelled sample dataset; achieved 88% test accuracy using TF-IDF features with a tuned Logistic Regression classifier.
- Built a complete MLOps pipeline with data and model versioning via MLflow and DVC; Dockerised the FastAPI inference service and deployed it on AWS EC2, enabling scalable real-time predictions with live monitoring.
- Developed a lightweight JavaScript Chrome extension that calls the prediction API on page load, surfacing comment-level sentiment badges directly within the YouTube UI.

Real Estate Properties Analytics Tool | **GitHub** *Jan 2024 - Mar 2024*

Python · XGBoost · Streamlit .DVC. Pandas · Scikit-learn

End-to-end predictive analytics platform for property price estimation and market trend visualisation.

- Built a 3-module platform covering price prediction, market analytics, and content-based property recommendations; structured as a reproducible pipeline using DVC for data versioning and experiment tracking.
- Developed an XGBoost regression model with engineered features (location encoding, bedroom/kitchen ratios, outlier-capped price bands), achieving 84% accuracy (R^2) on held-out test data.
- Deployed an interactive Streamlit dashboard with Plotly charts enabling buyers and sellers to explore market trends, compare listings, and receive top-5 property recommendations based on custom preferences.

CERTIFICATIONS & ACHIEVEMENTS

- Data Science Mentorship Program — CampusX(6-month structured program covering ML, EDA, and deployment)
- Ai Agents With Crew Ai — CampusX
- Ai Agents Using Agno — CampusX
- GenAI Using Ollama — CampusX
- Completed hands-on projects in Generative AI, MLOps, and LLM application development
- Built 4+ end-to-end AI/ML projects spanning NLP, recommendation systems, and analytics

EDUCATION

Bachelor of Science in Computer Science | University of South Asia *2021 – 2025*

Lahore, Pakistan